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Hazard Readiness AI System

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1. Introduction

1.1 Problem Statement

Disaster readiness, which is the capacity to predict hazards and navigate through emergencies, plays a vital role in minimizing potential harm and loss during sudden disaster outbreaks. Nowadays, the advancement in information technology allows systems to be developed to establish networks for sharing real-time situations during disasters to support effective decision-making. However, these systems have notable limitations that inhibit its flexibility and reliability. One of the limitations includes the lack of use of data provided by the public as there is no convenient and structured platform prepared for them to report hazards to the authority. Besides, privacy concerns may prevent the public from uploading the images or videos online because it may contain sensitive information such as their car license plates and faces. Last but not least, the routing system for evacuation may not keep track of real-time conditions and can lead users to unknown danger. Hence, a more comprehensive system is needed to enhance safety and orderliness during disaster evacuation.

1.2 Project Objectives

The main aim of this project is to build a web application that provides disaster readiness services by integrating crowdsourced disaster reporting, geospatial mapping and artificial intelligence (AI) interpretation. To be specific, the system developed is expected to offer medium for community hazard reports via photos, identify types of hazards using various AI-trained classification systems, implement privacy protection mechanisms such as blurring sensitive information in images, recommend fastest safe evacuation route based on real-time circumstances and alert the public when there is change in readiness level in a specific area. All of these objectives are to promote public situational awareness during disasters to avoid possible threats as well as possible.

1.3 UN-SDG Goals

This project also contributes to several United Nations Sustainable Development Goals (UN-SDGs). Firstly, it complements SDG 11: Sustainable Cities and Communities which

emphasizes on cities resilience and inclusivity. In this context, the disaster readiness web application helps boost urban safety and hazard consciousness by alerting hazard areas and preparing a secure evacuation route. Moreover, it supports SDG 13: Climate Action which aims to improve city capacity towards climate-related disasters. The implementation of spatial analysis allows forecasts of potential disasters and prepares the public for early adjustments.

2. Review on Existing Solution

2.1 Current Solution

Disaster management systems in Southeast Asia have improved significantly due to recurring floods, landslides, and severe weather events. In Malaysia, the National Disaster Management Agency (NADMA) leads national coordination through centralized data and communication systems, while the Malaysian Meteorological Department (MetMalaysia) provides weather forecasts and early alerts through its myCuaca app. These top-down systems deliver official warnings quickly but offer limited community-level detail.

At the regional level, platforms such as Google Crisis Map, UN-SPIDER, and the Global Disaster Alert and Coordination System (GDACS) integrate satellite data, global sensors, and humanitarian inputs to visualize disasters worldwide. Their broad coverage supports early detection but lacks the hyperlocal precision needed for specific neighborhoods or streets.

A useful regional model is PetaBencana, an open-source platform from Indonesia that crowdsources real-time flood updates through social media posts. While not active in Malaysia, PetaBencana demonstrates how community participation can complement official channels to improve ground awareness. Its approach serves as inspiration for developing localized, citizen-driven solutions that use modern data pipelines and AI for hazard verification.

Additionally, navigation apps like Waze and Google Maps allow Malaysian users to share road hazards and flooded routes during emergencies. Although not designed for disaster response, these examples show how participatory mapping can inform safer decision making when supported by location data and user interaction. Combined, these systems form a strong foundation for awareness and coordination, yet few focus on continuous community sensing or

adaptive, neighborhood-scale insights, an area the proposed Hyperlocal Disaster Readiness project aims to address.

2.2 Limitation of Current Solution

Existing disaster platforms remain limited by scale, data granularity, and integration. National systems like NADMA and MetMalaysia operate mainly at the regional level and rely on fixed sensors, leaving information gaps in smaller communities. Crowdsourced systems inspired by PetaBencana improve speed but rely on unstructured text or social content that may be inaccurate or incomplete.

Current tools also lack dynamic routing and real-time hazard adaptation. While navigation apps show temporary warnings, they do not calculate safe evacuation routes or factor in hazard severity from external data sources. Government systems, meanwhile, produce static maps that are not updated with live road or flood conditions.

Another constraint is interoperability. Data from various Malaysian agencies differ in formatting, update frequency, and classification. The lack of unified APIs or schemas makes it difficult to fuse hazard, traffic, and accessibility data into a single readiness score.

Finally, privacy and trust remain weak points. Some community-reporting platforms display user-submitted photos or locations without anonymization, risking exposure of identities, license plates, or private properties. Because many systems lack clear retention or consent policies, public confidence in sharing data can be limited.

In summary, Malaysia's current ecosystem provides strong national coordination but lacks hyperlocal precision, real-time adaptability, and assured data privacy. These gaps emphasize the need for an AI-supported, privacy-conscious platform that blends official data with community sensing to enhance local readiness and routing reliability.

3. Proposed Solution

3.1 System Overview

To address the weaknesses of existing disaster readiness systems, this project proposed a web-based disaster readiness application that combines community-based hazard data collection, AI classification, geospatial analysis and adaptive routing to provide the most relevant guidance for users to navigate through dangerous environments during disasters. This system can be accessed through mobile devices and desktop browsers, where users can interact with the displayed map that shows information on locations of hazards, shelters, evacuation paths and readiness scores of each grid. Users are provided with hazard report submission services where they can upload images for the system to detect and interpret their positions on the map through Global Positioning System (GPS). These reports are then sent to the system backend for further processing by the AI pipeline, which has two main functions. First, it predicts the hazard types shown in the uploaded images and assigns a confidence score to each of them. Then, a privacy protection mechanism is activated to automatically blur sensitive information in the images such as vehicle license numbers.

After that, based on the hazard identified, the system will conduct geospatial analysis within a defined radius around the hazard location to update the risk penalty of the roads nearby and adjust the readiness score of the surrounding grid cells. This readiness score also takes into consideration local baseline vulnerability, recent hazard records and accessibility to nearby shelters. When the readiness score falls below a certain critical threshold, the system will generate alerts on the dashboard to signal the users to be aware. Finally, users can search for the fastest evacuation path by inputting their current location and selecting the desired shelter as their destination. The routing engine will then evaluate all possible paths based on latest hazard parameters and suggest the route with the least estimated time of arrival (ETA) to guide the users to safety as soon as possible.

3.2 Key Features

The following are the key features that are expected to be functional within the system:

1. Community-based Hazard Reporting

Users can help provide hazard information they personally encounter by uploading images taken to feed more data to the system which can also help to improve its performance in the long run.

2. AI Hazard Classification

AI analysis helps to decipher and classify the hazards shown in the images received into categories such as normal conditions, floods, fire and landslides and provide confidence scores to indicate reliability of the predictions.

3. Image Redaction for Privacy Preservation

Information that users are likely reluctant to reveal such as their faces and their vehicle license plates is automatically blurred before storage and display. This reassures users to submit reports without fear of being exposed to information leaks.

4. Automatic Alert Prompt

Abnormally low readiness scores that indicate serious hazards will spontaneously trigger alerts that highlight the locations involved. This allows users to acknowledge the potential threats as soon as possible and make quick decisions.

5. Adaptive Routing Mechanism for Effective Evacuation

The routing engine developed accounts for hazard parameters such as risk penalty to reassess feasibility of paths affected by hazards. The risk penalties are also continuously updated according to new hazard reports to make timely adjustments to the routing system.

4. Methodology

4.1 System Development Approach

The system was developed as a modular monorepo structure, where each major capability was implemented as an independent component with clear interfaces, then integrated in stages.

The development process used progressive backend switching, where early development used mock services to unblock frontend and endpoint wiring, followed by migration to SQL-backed services once the PostGIS/pgRouting schema and readiness functions were stable. This allowed rapid validation of user flows before full database coupling. The same strategy was applied to asynchronous processing, where report image handling was first modeled via queue contracts and then consolidated into an in-process background pipeline inside the API container for operational simplicity.

System contracts were defined via REST endpoints under /api/v1, with schema validation through Pydantic models to ensure consistent payloads between frontend and backend. Geospatial and readiness logic were moved into database functions to centralize computation close to the data layer and minimize duplicated scoring logic in application code.

Quality control combined unit level API tests, integration checks for SQL/routing adapters, and scenario-based verification of readiness updates and alert triggers. Security and safety controls were included, including upload validation, rate limiting, and privacy-preserving image handling. This development approach balanced speed, modularity, and reliability, and enabled each subsystem to be verified independently before full end-to-end orchestration.

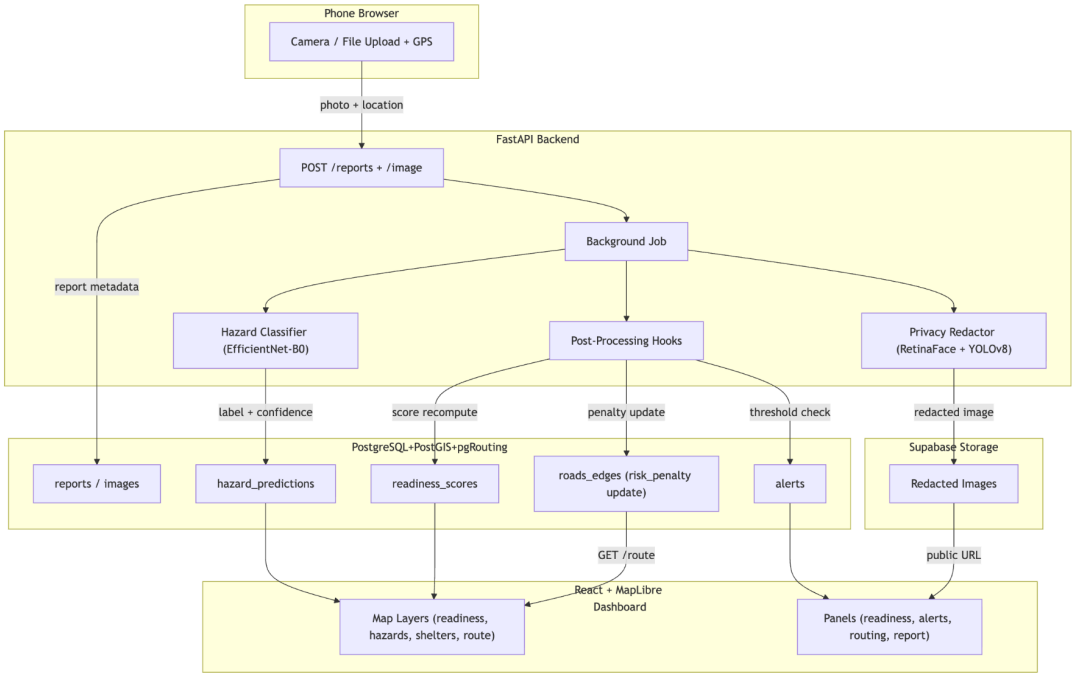

Figure 4.1.1 System Architecture

4.2 Data Collection & Preprocessing

4.2.1 Hazard Classification Model

The hazard image dataset used for training the classification model was compiled from Kaggle and Roboflow repositories (Karim, 2023; Tandulje, 2025; Roboflow, 2025). The dataset contains four hazard categories, each labeled according to its corresponding class. A total of 2,110 images were collected to create a balanced dataset across all labels. The dataset consists of RGB images in .jpg format. The diversity of this dataset makes it suitable for training a hazard classification model for real-world scenarios.

Before training, images were resized to 224×224 pixels to standardize the input size for the classification model. Data augmentation techniques including random horizontal flipping, random rotation up to 15°, and color jitter adjustments were applied. These transformations were crucial to simulate the varying lighting conditions and camera angles commonly found in real-world conditions. For the validation dataset, only resizing and normalization were applied to ensure consistent model evaluation without additional augmentation.

4.2.2 License Plate Object Detection Model

The dataset used for our license plate detection model training was retrieved from the License Plate Malaysia dataset on Roboflow repository (Plate, 2025). The dataset was manually annotated with bounding boxes to identify Malaysian license plates captured under varying lighting conditions, viewing angles, and environmental settings. Therefore, these annotations make it suitable for our Malaysia license plate object detection model training.

For preprocessing, images were automatically oriented and resized to 640×640 pixels to standardize input dimensions for the YOLOv8 detection model. Data augmentation techniques including horizontal and vertical flipping, random rotation up to 15° , shear transformations up to 10°, grayscale conversion applied to 15% of images, and brightness adjustments were used to improve model generalization across varying environmental conditions.

4.2.3 Geospatial

The geospatial dataset in this project originated from free open sources online. The main road network dataset for Petaling Jaya, Petaling, Selangor, Malaysia was retrieved from OpenStreetMap (OSM) before converting into a graph structure that is more ideal for dynamic routing. The sample shelter locations such as schools and community centres within Petaling Jaya were also chosen from the collection of nodes extracted from OSM. In addition, the 30m resolution Digital Elevation Model (DEM) of Petaling Jaya was downloaded from OpenTopography in HGT format to provide sufficient terrain information for hazard analysis. Last but not least, the dataset of historical flood zones were gathered based on online reports through news and social media to provide past context for more accurate readiness scoring.

4.3 Hazard Classification Model Development

EfficientNet-B0 convolutional neural network was used for our hazard classification model. This selection was made due to its capability of achieving high accuracy while having a small model size of estimated 6.06 million parameters, which is relatively fewer than other conventional CNN models (Kurniawan et al., 2025). Therefore, this lightweight design with

lower computational cost enables faster training, making it suitable for our web-based system and mobile-friendly environment where performance efficiency is an important consideration.

Additionally, the transfer learning approach was applied by initializing the EfficientNet-B0 model with the pretrained weights from the ImageNet dataset. This is crucial in improving the model performance when training on smaller and task-specific datasets by allowing our model to reuse visual features learned from the large-scale image datasets. Furthermore, the final classification layer was then modified to output predictions for the four hazard classes (fire, flood, landslide, normal) used in this project.

During the model training, it was conducted using the Adam optimizer with a learning rate of 0.001 and Cross Entropy Loss function for multiclass classification. Moreover, the model was trained for a maximum of 30 epochs, and an early stopping mechanism with a patience of 5 epochs was implemented to prevent overfitting. This mechanism was crucial in monitoring the validation loss and stops the training process when it does not improve over five consecutive epochs. The best performing model was then saved and evaluated on the test set based on validation loss. We have also assessed our model performance using classification accuracy, precision, recall, and F1-score, along with a confusion matrix to analyze the classification performance across all hazard classes.

Finally, the trained model will produce a predicted hazard label and its corresponding confidence score which represents the probability of the predicted class. This model is used by the disaster readiness system to automatically classify hazard reports submitted by users. Moving on, these outputs will then be integrated into the system to support automated hazard reporting, and update the hazard information on the map such as rerouting adjustments and readiness score.

4.4 Privacy Redaction Pipeline Implementation

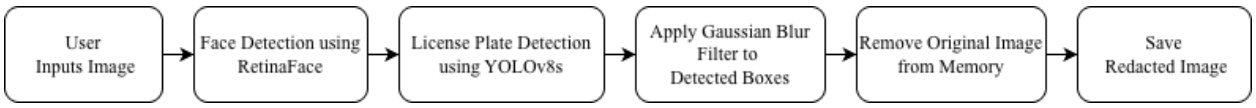

Figure 4.4.1 Privacy Redaction Pipeline

To maintain users confidentiality when uploading hazard images, we have implemented a privacy redaction pipeline as shown in Figure 4.4.1 in the image processing workflow that automatically detects and blurs sensitive visual information before proceeding with further analysis. Original unredacted images are also removed from memory after processing, to ensure only the redacted image is retained in the system. To be precise, the pipeline focuses on hiding personally identifiable information such as human faces and vehicle license plates.

Face Detection Model

RetinaFace deep learning framework was selected for our face detection model. This was supported by its high accuracy in localizing faces even under challenging scenarios such as multiple small faces, occlusions, and different lighting conditions (Mahamad Amin et al., 2025). Since the model is pretrained on a large scale facial dataset, we can directly apply it without additional training. To filter our low confidence detections and reduce false positives, a confidence threshold of 0.6 was applied. Finally, the model outputs bounding boxes that represent the face regions in the image.

Vehicle License Plates Detection Model

A custom model with annotated dataset images of Malaysian license plates with bounding boxes was trained using the YOLOv8 object detection architecture. Moreover, the YOLOv8s variant was selected due to its balance between detection accuracy and computational efficiency (Ultralytics, n.d.). During model training, an image resolution of 640x640, batch size of 8, and up to 100 epochs was utilized. An early stopping mechanism with patience of 20 epochs was also applied to prevent overfitting and the best performing model was saved. Most importantly, the model performance was monitored using the mean Average Precision (mAP@50) to measure the model's ability in correctly localizing and detecting license plates.

Gaussian Blur Filter

Once the face and license plate model identifies the sensitive regions, the detected bounding boxes will then be passed to the redaction module. The redaction process applies a Gaussian blur filter to the detected regions to blur the confidential information (Team, 2016). Moving on, each detected region will then be extracted from the original image, and blurred

using a strong kernel size of 31x31, and then replaced back into the image to produce the final redacted output image.

4.5 Geospatial Data Processing

Geospatial data processing is a core component of the disaster readiness prototype that is responsible for analysing spatial associations between hazards, shelters, road networks and neighbourhood readiness levels. In this context, the spatial database used in the system was built on PostgresSQL with the PostGIS extension for efficient storage of geographical data.

The geospatial data processing pipeline began with the extraction of the road map of the selected study area, which is Petaling Jaya, Petaling, Selangor, Malaysia from the open source OpenStreetMap (OSM). Then, the dataset was imported into PostgresSQL and was transformed into a routable graph structure with each road segment represented as edge with corresponding attributes such as length and cost. Additionally, GiST indexes were incorporated to improve speed of geospatial queries. Moreover, several complementary geospatial layers were also injected into the database, including administrative boundaries, shelter locations, grid cells and neighbourhoods used for readiness scoring. This method of dividing the map into grid cells allows computation of local readiness score to track spatial data changes in specific areas.

After a user submits a hazard report in the form of images, the system will capture the target coordinates through GPS or pin drop on map. The spatial queries implemented will then decide which grid cell and neighbourhood the disaster occurs at and adjust the readiness score based on the hazard type. Furthermore, spatial analysis using proximity query was also applied to identify road segments that are affected by the hazard within a certain defined radius. These affected road segments are assigned with higher risk penalties which influences the calculation of the fastest secure evacuation route.

In addition, past hazard records are also taken into consideration and are aggregated over time to extract spatial patterns. For instance, a grid cell with multiple disasters reported within it is associated with relatively higher vulnerability. Besides, more recent hazards are weighted more heavily by the AI classification model using confidence scores and temporal filtering. In brief, the geospatial data processing pipeline that integrated spatial indexing, proximity analysis

and hazard aggregation allows the system to update the routing system based on real-time and localized readiness score to facilitate more precise decision-making during disasters.

4.6 Routing Algorithm

Unlike traditional navigation systems, the routing system of the prototype developed within this project implemented adaptive risk calculation based on hazard reports to optimize travel time of recommended evacuation path. The edges within the road graph are designated with a base cost that represents ETA of the corresponding road segment in normal condition. To account for hazard events, an additional parameter named risk penalty was introduced to adjust the cost according to current situations to avoid dangerous paths. As an illustration, roads around the reported flooded areas are given higher penalties, increasing the final edge costs and making them less suitable for route selection. The final cost is hence computed as the sum of the base cost and risk penalty for each road segment. Since the system will be continuously fed with new hazard data, the risk penalties are constantly changing and allows dynamic routing that nearly simulates real-time updates to provide guidance that best reflects the latest situations to users.

Furthermore, the system used pgRouting extension of PostgresSQL to utilize shortest-path algorithms such as Dijkstra's algorithm or A* search to determine the most optimal path in terms of time between location of the user and the shelter. Both of these algorithms will consider all possible driveable routes and select the one with the lowest final cost. In addition, the system also performs a snapping mechanism before route computation to map both the user's and the shelter's real geographic coordinates to the nearest nodes established in the road graph generated. This helps to guarantee the validity of the starting and ending point for smooth routing procedures.

Finally, the resultant route is then exported to a GeoJSON format and returned to the frontend application. Route polyline layer renders are then used to visualize the route on the map along with labelled information such as distance and ETA. By combining graph-based routing algorithms and flexible hazard penalty adjustments, the system offers a more reliable and accurate navigation system during disasters to ensure safety of users from time to time.

4.7 Risk/Imputation Model for Hazard Readiness

The risk/imputation model estimates baseline vulnerability for each analysis grid cell in Petaling Jaya, particularly in areas with limited historical damage labels. Its purpose is to produce a continuous risk prior in the range of 0 to 1 for downstream readiness scoring. Features were extracted from terrain and exposure datasets, including elevation and slope (SRTM), distance to waterways (OSM), distance to known flood hotspots, and accessibility signals such as road density and shelter travel time.

Because accurate large-scale ground-truth labels were unavailable, the model was trained on a constructed proxy target defined as `vulnerability_proxy = 0.50*(hotspot_proximity_score) + 0.25*(low_elevation_score) + 0.25*(river_proximity_score)`. All three components are normalized to the same scale before being combined, so that areas closer to historical hotspots, lower in elevation, or nearer to waterways receive higher vulnerability estimates. An XGBoost regressor was selected for its ability to capture nonlinear relationships between the geospatial input features, using a baseline configuration of 200 trees, a maximum depth of 5, a learning rate of 0.1, and a subsampling ratio of 0.8. The resulting per cell vulnerability scores are bounded between 0 and 1 and stored in the database as each cell's baseline vulnerability, where they serve as the structural risk component in the readiness scoring model.

Hazard readiness is computed per cell with a composite SQL-based scoring function that combines short term hazard activity, baseline vulnerability, accessibility, and data confidence. The pipeline aggregates recent hazards, with default set to a 24-hour window, then joins baseline vulnerability from the imputation model and average travel time to shelter from accessibility data. The score is constrained to 0 to 100 by starting from 100, subtracting hazard and vulnerability penalties, and adding accessibility and confidence bonuses. Coverage confidence is modeled with `coverage = min(1, hazard_count / 20)`, which both informs the confidence bonus and is stored for interpretability in low-report areas. Each update persists the final score alongside a detailed breakdown that records the hazard and vulnerability penalties, the accessibility and confidence bonuses, the number of recent hazard reports, a time decay factor, and the current observation coverage. Alert rules are then evaluated, with low-readiness

thresholds and severe-hazard probability checks generating high-severity alerts when conditions are met.

4.8 Backend API and System Integration

The backend was implemented in FastAPI with versioned endpoints under /api/v1 and a layered architecture that consists of routes, services, repositories, and ORM models. Integration is contract driven so frontend, AI processing, routing, and readiness modules could be developed and swapped independently without changing client facing payloads. The API handles report creation, image upload, asynchronous processing orchestration, hazard/readiness/alert retrieval, route computation, and health checks.

In the core ingestion pipeline, a client submits report metadata and then uploads an image, after which the backend validates file constraints and applies endpoint rate limiting. The image is processed in the background for hazard classification and privacy redaction, the redacted output is stored in Supabase Storage, and prediction plus image metadata are persisted to PostgreSQL. Post-processing hooks then update nearby road penalties, recompute readiness for affected cells, and evaluate alert triggers. Spatial cell resolution, readiness scoring, and alert logic are handled through database functions, while the routing engine uses road network tables whose traversal costs are dynamically updated as new hazard events are reported.

Dependency injection and environment based backend selection are used to support development and deployment flexibility, where mock or real environments can be chosen while preserving stable API contracts. Operational safeguards include CORS controls, typed request validation, anti-spam throttling, upload size/type checks, and privacy guardrails that expose only sanitized redacted image URLs. Together, this forms an end-to-end flow from community report submission to frontend displayed hazards, readiness scores, alerts, and evacuation routes.

4.9 Frontend Interface Development

The main objective of the frontend interface is to provide an intuitive and interactive platform for users to visualize complex hazard related information easily with user-friendly

accessibility. To achieve this goal, modern web technologies and mapping libraries were used to ensure high performance, responsiveness, and scalability.

The application was implemented using React, which enables the development of modular and reusable interface components. React's component-based architecture allows different parts of the interface to be organized logically, improving maintainability and scalability. Additionally, React supports efficient rendering through its virtual DOM mechanism, ensuring that updates to the interface remain responsive even when handling dynamic data (Banks & Porcello, 2020).

For geographic visualization, the system integrates MapLibre GL JS, an open-source library for rendering interactive vector maps directly in the browser using WebGL technology. The basemap tiles and styling are provided through MapTiler via API, which supplies high-quality mapping services suitable for web applications. This combination enables the application to deliver smooth and efficient map interactions while supporting the visualization of spatial hazard data.

The interface is structured around an interactive map component, which serves as the central element of the application. Surrounding the map are several functional panels designed to organize different types of information for the user. These panels include the Hazard panel, Readiness panel, Alerts panel, and Route panel. Each panel provides specific information relevant to disaster awareness and response. For instance, the Hazard panel displays hazard-related data, while the Readiness and Alerts panels provide guidance and important notifications. The Route panel allows users to view navigation-related information that may assist in evacuation or safe movement during emergencies. By separating information into dedicated panels, the interface ensures that users can easily access relevant content without overwhelming the main visualization area.

Another essential component of the frontend is the hazard reporting functionality, which allows users to contribute real-time information to the system. Users can report hazards by uploading images directly through the interface. The reporting process requires users to capture or upload a photo, select the corresponding geographic location on the map, and confirm their submission by checking a consent checkbox before the data is sent to the database. This

workflow ensures that user-submitted information is collected responsibly while maintaining transparency regarding data usage.

The map interface was designed to be highly interactive to improve user engagement and usability. Most elements displayed on the map are clickable, allowing users to retrieve additional information through interactive pop-ups or interface responses. This interactivity helps users explore spatial information more effectively and gain deeper insight into hazard-related data.

In addition, the system includes a map legend with layer controls, which enables users to toggle specific layers on or off. This feature allows users to customize the map view according to their needs, reducing visual clutter and improving clarity when multiple layers are present. By allowing users to selectively display information, the interface provides greater flexibility and enhances the overall user experience.

Overall, the frontend interface combines modern web technologies with interactive geospatial visualization to create a responsive and user-friendly system. The integration of multiple information panels, hazard reporting capabilities, and customizable map controls ensures that users can both consume and contribute information effectively.

5. Results

5.1 Outcomes

The system met its core objectives, delivering a functional end-to-end disaster readiness platform covering community hazard reporting, AI-driven image processing, dynamic routing, and live readiness scoring for the Petaling Jaya study area.

The hazard classification model achieved a test accuracy of 99.5% and a macro F1-score of 0.996 across four hazard classes, fire, flood, landslide, and normal, with per-class precision and recall at or near 1.0 for all categories. The privacy redaction pipeline for license plate detection achieved a mean average precision of 0.83 at the standard IoU threshold, providing reliable automated blurring of identifiable information before any image is stored or displayed. The vulnerability imputation model produced highly consistent estimates against its proxy labels, with a mean absolute error of 0.013 and an R^2 of 0.994, indicating strong fit across the 459 grid cells covering the study area.

At the system level, readiness scores were successfully computed and stored for all 66 neighbourhoods in Petaling Jaya, achieving full spatial coverage. The routing engine was seeded with 18,426 road segments and verified to return valid evacuation paths that reroute dynamically as hazard events raise road traversal costs. The full report-to-readiness pipeline, from image upload through classification, redaction, road penalty update, and score recomputation, was verified end-to-end via integration tests, confirming that all post-processing hooks fire correctly and that readiness breakdowns and alerts are generated as expected. The frontend successfully renders all neighbourhood readiness scores, hazard markers, shelter pins, and computed routes from live API data.

A showcase of the prototype is shown below, with additional screenshots being provided in the GitHub repository's README.md.

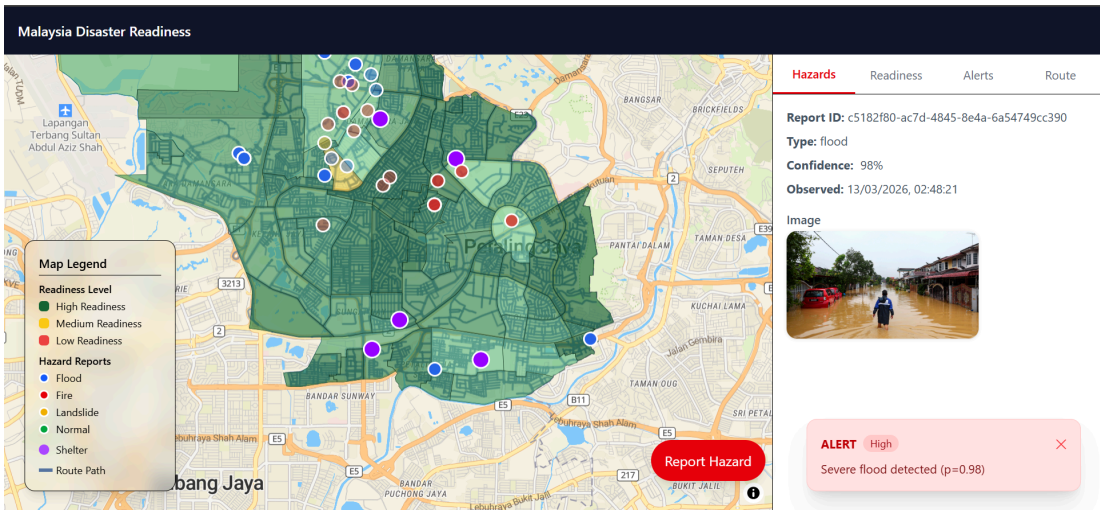

Figure 5.1.1 UI Screenshot of prototype

6.0 AI Usage Acknowledgement

AI tools were used as support tools during development and documentation. ChatGPT was used during project planning and brainstorming phases, and also to refine report writing and improve clarity of wording, while Gemini was used to assist with code debugging and troubleshooting during implementation.

6. Conclusion

7.1 Strength & Limitation

Our proposed Hazard Readiness AI System has the ability to combine community participation, artificial intelligence, and geospatial intelligence into a single platform. The system allows citizens to report hazards by uploading images, which are automatically analyzed using an AI classification model up to 99.53% accuracy. Moreover, our system also considered the user's privacy by implementing a privacy redaction pipeline which blur faces and vehicle license plates. This system ensures the user's confidentiality and encourages people to share useful information safely. In addition, the system provides real-time hazard mapping and dynamic evacuation routing, which allows users to better understand the situation around them and make safer decisions during emergencies. Furthermore, we have also included an area readiness score which helps the users to have a disaster risk levels summarization within a specific location.

However, the system solely relies on the user's reported submission. Thus, areas with lower submission may have limited data coverage. Furthermore, the routing and readiness scoring functions depend on the available geospatial and road data. Therefore, it may not always be updated with sudden changes during disasters. As the system is currently developed as a prototype, further testing and integration with official disaster management agencies would be necessary for a more practical deployment.

7.2 Future Improvements

For further enhancements, expanding the training dataset to include more disaster categories would make our system more diverse. Moreover, integrating additional real-time data sources such as weather information, flood sensors, or satellite data could further improve hazard detection. In addition, we can also include live traffic data which can enhance the reliability of evacuation routes. Finally, we may improve user engagement through developing mobile-friendly interfaces and notification systems that could encourage more people to contribute hazard reports, increasing the system's overall effectiveness in supporting disaster readiness.

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